

Assignment 3

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Importing necessary packages:

```
library(readr)
library(tidyr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(pastecs)
```

```
##
## Attaching package: 'pastecs'

## The following objects are masked from 'package:dplyr':
##
##   first, last

## The following object is masked from 'package:tidyr':
##
##   extract
```

Importing the dataset (TextMessages):

```
TextMessages <- read_csv("TextMessages.csv")
```

```
## Rows: 50 Columns: 4
## -- Column specification -----
## Delimiter: ","
## dbf (4): Group, Baseline, Six_months, Participant
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
names(TextMessages) # shows variable names
```

```
## [1] "Group"      "Baseline"    "Six_months"  "Participant"
```

```
head(TextMessages) # shows first 6 rows
```

```
## # A tibble: 6 x 4
##   Group Baseline Six_months Participant
##   <dbl>   <dbl>   <dbl>         <dbl>
## 1     1     52     32             1
## 2     1     68     48             2
## 3     1     85     62             3
## 4     1     47     16             4
## 5     1     73     63             5
## 6     1     57     53             6
```

Renaming Group variable for easier read:

```
TextMessages <- TextMessages %>%
  mutate(Group = case_when(
    Group == 1 ~ "Group 1",
    Group == 2 ~ "Group 2"
  ))
head(TextMessages$Group)
```

```
## [1] "Group 1" "Group 1" "Group 1" "Group 1" "Group 1" "Group 1"
```

Converting data from wide to long format: This allows for Time to be represented in two categories: baseline and six_months. It takes those values and assigns it to that specific time period. Now, this long data set has the variables Group, Participant, Time, and Messages.

```
TextMessages_long <- TextMessages %>%
  pivot_longer(
    cols = c(Baseline, Six_months),
    names_to = "Time",
    values_to = "Messages"
  )
names(TextMessages_long)
```

```
## [1] "Group"      "Participant" "Time"         "Messages"
```

```
head(TextMessages_long)
```

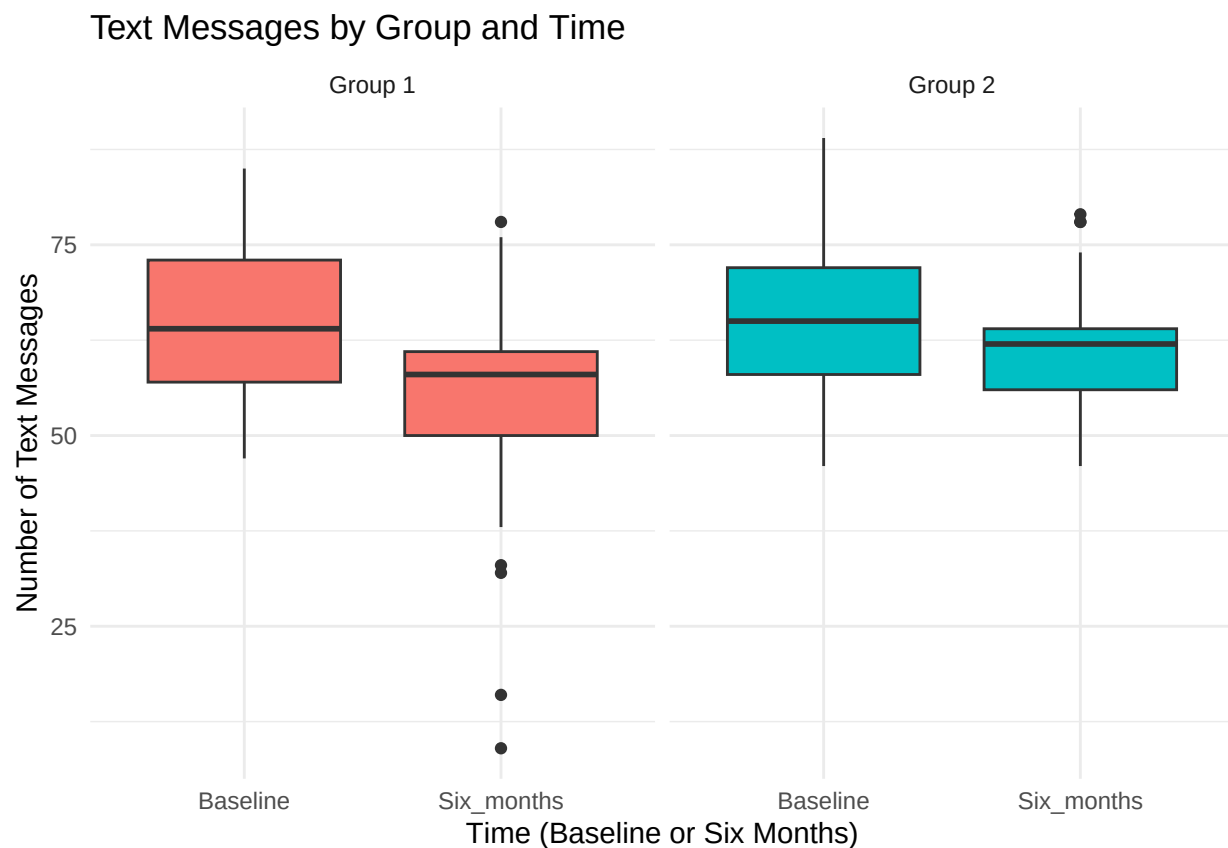
```
## # A tibble: 6 x 4
##   Group Participant Time      Messages
##   <chr>         <dbl> <chr>         <dbl>
## 1 Group 1           1 Baseline         52
## 2 Group 1           1 Six_months        32
```

## 3	Group 1	2 Baseline	68
## 4	Group 1	2 Six_months	48
## 5	Group 1	3 Baseline	85
## 6	Group 1	3 Six_months	62

Visualization #1 - Bryanna: Stratified boxplots of text messages by Group and Time

Below, we used the long format dataset to show text messages by group and time. To do so, we filled and wrapped our plot with the Group variable (Group 1 and Group 2). We see a similar trend of text messages over time for each group. There appears to be a decrease in the median number of text messages from baseline to six months after the baseline. However, it appears that Group 2 has a greater median number of text messages overall, meaning at the baseline and six months after. There are multiple outliers at the six month post baseline for group 1, while for group 2 there appears to only be two outliers.

```
ggplot(TextMessages_long, aes(x = Time, y = Messages, fill = Group)) + geom_boxplot() +
  facet_wrap(~ Group) +
  labs(
    title = "Text Messages by Group and Time",
    x = "Time (Baseline or Six Months)",
    y = "Number of Text Messages"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

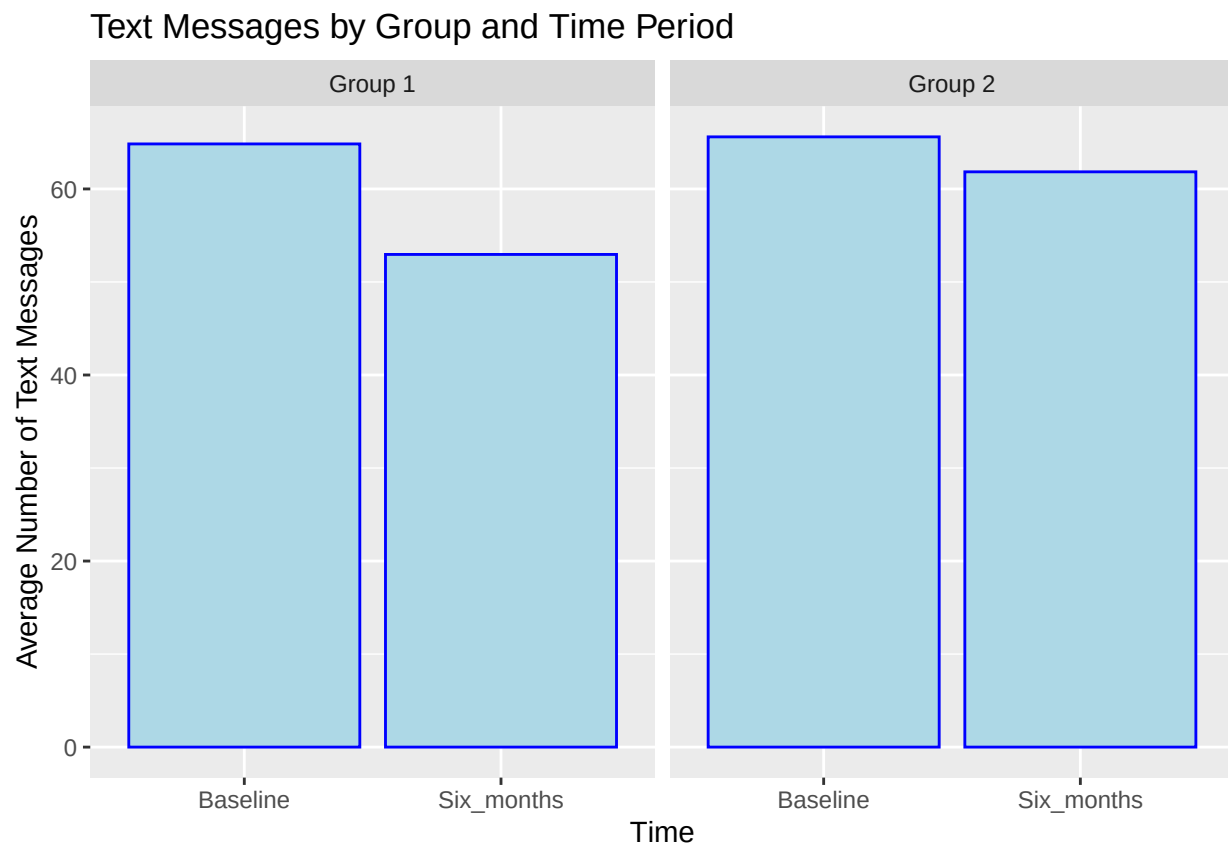


Visualization #2 - Ibrahim: Stratified bar plot of text messages by Group and Time

The bar plot shows difference in means between baseline and six months after the baseline period for each group. According to the plot, in this sample, the average number of text messages is significantly higher in

the baseline than six months after for both groups (Group 1 and Group 2). The average number of text messages in the baseline seems same in Group 1 and Group 2, while in six month after the average is lower in Group 1 than it in Group 2.

```
barplot_d_by_Group_and_Time<-ggplot(TextMessages_long,aes(x = Time,
                                                             y = Messages, fill = Group))
barplot_d_by_Group_and_Time+
  stat_summary(fun.data = mean_cl_normal,geom = "bar", colour = "blue" ,
              fill="lightblue") +
  labs(title="Text Messages by Group and Time Period",
       x="Time",
       y="Average Number of Text Messages") +
  facet_wrap(~Group, nrow = 1)
```



Summary Statistics - Bryanna: Summary statistics of Text Messages by Group and Time. I also performed summary statistics of Text Messages by Group and Text Messages by Time.

Based on the summary statistics of text messages by both group and time, we see the greatest mean text messages to be 65.6 in the baseline measurement of group 2. We see the minimum value of text messages in group 1 six months after the baseline is 9, which is seen as an outlier in the boxplot above. Group 1 six months after the baseline has the greatest amount of variation of approximately 266.71 with a range of 69, while group 2 six months after the baseline has the least amount of variation of 88.56 with a range of 33. This also follows the greatest standard deviation of 16.33 in group 1 six months after baseline, while for group 2 six months after baseline the standard deviation is 9.41.

```
by(TextMessages_long$Messages, TextMessages_long$Group, stat.desc)
```

```
## TextMessages_long$Group: Group 1
##      nbr.val    nbr.null    nbr.na      min      max      range
##  50.0000000    0.0000000    0.0000000    9.0000000   85.0000000   76.0000000
##      sum      median      mean    SE.mean CI.mean.0.95      var
## 2945.0000000   60.0000000   58.9000000    2.1095023    4.2392036  222.5000000
##      std.dev    coef.var
##   14.9164339    0.2532502
## -----
## TextMessages_long$Group: Group 2
##      nbr.val    nbr.null    nbr.na      min      max      range
##  50.0000000    0.0000000    0.0000000   46.0000000   89.0000000   43.0000000
##      sum      median      mean    SE.mean CI.mean.0.95      var
## 3186.0000000   62.0000000   63.7200000    1.4456239    2.9050901  104.4914286
##      std.dev    coef.var
##   10.2221049    0.1604222
```

```
by(TextMessages_long$Messages, TextMessages_long$Time, stat.desc)
```

```
## TextMessages_long$Time: Baseline
##      nbr.val    nbr.null    nbr.na      min      max      range
##  50.0000000    0.0000000    0.0000000   46.0000000   89.0000000   43.0000000
##      sum      median      mean    SE.mean CI.mean.0.95      var
## 3261.0000000   64.5000000   65.2200000    1.5067982    3.0280244  113.5220408
##      std.dev    coef.var
##   10.6546723    0.1633651
## -----
## TextMessages_long$Time: Six_months
##      nbr.val    nbr.null    nbr.na      min      max      range
##  50.0000000    0.0000000    0.0000000    9.0000000   79.0000000   70.0000000
##      sum      median      mean    SE.mean CI.mean.0.95      var
## 2870.0000000   60.5000000   57.4000000    1.9703931    3.9596532  194.1224490
##      std.dev    coef.var
##   13.9327832    0.2427314
```

```
summary <- by(TextMessages_long$Messages,
  list(Group = TextMessages_long$Group, Time = TextMessages_long$Time),
  stat.desc)
```

```
summary
```

```
## Group: Group 1
## Time: Baseline
##      nbr.val    nbr.null    nbr.na      min      max      range
##   25.000000    0.000000    0.000000   47.000000   85.000000   38.000000
##      sum      median      mean    SE.mean CI.mean.0.95      var
## 1621.000000    64.000000   64.840000    2.135946    4.408377  114.056667
##      std.dev    coef.var
##   10.679732    0.164709
## -----
```

```

## Group: Group 2
## Time: Baseline
##      nbr.val      nbr.null      nbr.na      min      max      range
## 25.0000000  0.0000000  0.0000000  46.0000000  89.0000000  43.0000000
##      sum      median      mean      SE.mean  CI.mean.0.95      var
## 1640.0000000  65.0000000  65.6000000  2.1671794  4.4728385  117.4166667
##      std.dev      coef.var
## 10.8358971  0.1651814
## -----
## Group: Group 1
## Time: Six_months
##      nbr.val      nbr.null      nbr.na      min      max      range
## 25.0000000  0.0000000  0.0000000  9.0000000  78.0000000  69.0000000
##      sum      median      mean      SE.mean  CI.mean.0.95      var
## 1324.0000000  58.0000000  52.9600000  3.2662313  6.7411700  266.7066667
##      std.dev      coef.var
## 16.3311563  0.3083678
## -----
## Group: Group 2
## Time: Six_months
##      nbr.val      nbr.null      nbr.na      min      max      range
## 25.0000000  0.0000000  0.0000000  46.0000000  79.0000000  33.0000000
##      sum      median      mean      SE.mean  CI.mean.0.95      var
## 1546.0000000  62.0000000  61.8400000  1.8820910  3.8844450  88.5566667
##      std.dev      coef.var
## 9.4104552  0.1521742

```